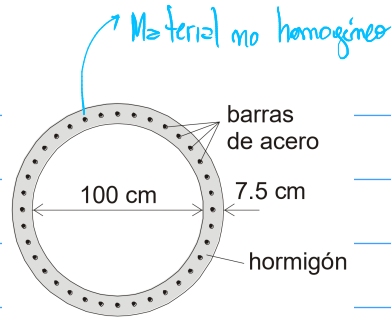
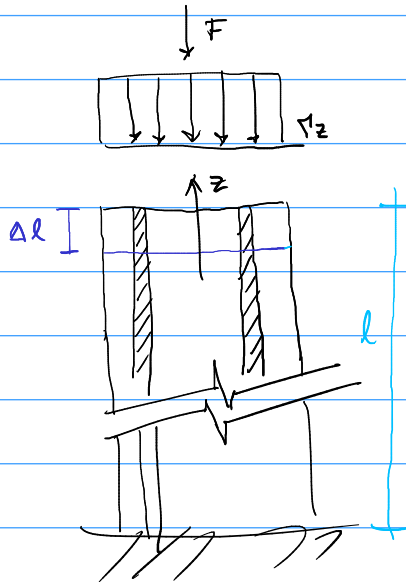


$$\nu_a = 0,25 \text{ y } \nu_h = 0,4 \text{ no se usa}$$



$$\frac{E_a}{E_h} = 15 = \alpha$$



$$\Delta l_a = \Delta l_h$$

$$l_a = l_h$$

$$\epsilon_{zz} = \epsilon_{zza} = \epsilon_{zzh} = \frac{\Delta l_a}{l_a} = \frac{\Delta l_h}{l_h} = \frac{\Delta l}{l}$$

$$\epsilon_{zz} = \frac{1}{E} [\sigma_{zz} - \nu (\sigma_{xx} + \sigma_{yy})]$$

→ no despreciables (se considera que no hay restricciones a la expansión lateral, tanto del acero como en el hormigón)

$$\epsilon_{zz} = \frac{\sigma_{zz}}{E}$$

$$\epsilon_{zza} = \frac{\sigma_{zza}}{E_a}$$

$$\sigma_{zza} = \frac{F_a}{A_a}$$

$$\epsilon_{zzh} = \frac{\sigma_{zzh}}{E_h}$$

$$\sigma_{zzh} = \frac{F_h}{A_h}$$

$$F = F_a + F_h$$

$$\epsilon_{zza} = \frac{F_a}{A_a E_a}$$

$$\epsilon_{zzh} = \frac{F_h}{A_h E_h}$$

$$F_a = F_a \left(1 + \frac{F_h}{F_a}\right)$$

$$F_h = F - F_a$$

Al ser iguales las deformaciones

$$\frac{F_a}{A_a E_a} = \frac{F_h}{A_h E_h} \Rightarrow \frac{F_a}{F_h} = \frac{A_a E_a}{A_h E_h}$$

$$F_a = \frac{1}{1 + \frac{F_h}{F_a}} F = \frac{1}{1 + \frac{A_h E_h}{A_a E_a}} F = \frac{A_a E_a}{A_a E_a + A_h E_h} F$$

$$F_h = (1 - \alpha) F = \frac{A_h E_h}{A_a E_a + A_h E_h} F$$

$$\frac{A_h E_h}{A_e E_e + A_h E_h} = \frac{A_h E_h}{E_h (A_e \frac{E_e}{E_h} + A_h)} = \frac{A_h}{A_e \gamma + A_h}$$

$$A_h = \pi (r_e^2 - r_i^2) - A_a$$

$$A_a = 36 \times 6,5 \text{ cm}^2$$

$$\frac{A_a E_a}{A_a E_a + A_h E_h} = \frac{E_a}{E_h} \frac{A_a}{(A_e \frac{E_e}{E_h} + A_h)} =$$

$$= \frac{\gamma A_a}{A_e \gamma + A_h}$$